

Final Term Project: Law, Logic and LLMs

The Objective: Using SMT for Guardrailing LLMs

This final project replaces a written exam and builds on the concepts and techniques you developed in earlier projects. This project focuses on applying guardrails to AI systems. Your task is to train a language model to decide whether an individual qualifies as a "dependent" under [26 USC 152\(a\)-\(d\)](#), the same provision you analyzed in Projects 4 and 5. After training, you will evaluate your model using the SMT-LIB formalization you constructed in the previous assignments.

You must complete this project on your own. Teams are not allowed, but you can use the formalization from your previous assignments, only if it was written by you. Because this project serves as a substitute for a final exam, you cannot discuss the project and the solution with your peers. You also cannot ask questions about the project on Ed, in office hours, or by writing to the course staff, unless it is a clarifying question about parts of the project description that are not clear. Requests for library installations on the HPC are allowed as well. (If you do ask something about the final project on Ed, the post should be private.)

Part 1: Create Your Dataset

Construct a training dataset to teach the model how to assess whether an individual qualifies as a dependent. Your dataset should reflect relevant legal information about dependency status (e.g., e.g., [statutes](#), prompt-response pairs, [IRS publications](#), etc).

Dataset Size Ablation

Rather than choosing an arbitrary dataset size, you will construct datasets of **three different sizes** and evaluate how size affects your model's downstream performance:

- **Small:** 10 examples
- **Medium:** 20 examples
- **Large:** 50 examples

Train a separate model (or checkpoint) on each dataset size. This ablation is part of your evaluation in Part 3.

DPO Format

Because you will be using Direct Preference Optimization (see Part 2), your dataset must include **preference-formatted examples**. Each example must follow this exact template:

```
{  
  "prompt": "<a fact pattern or legal question about dependency status>",  
  "chosen": "<a correct, well-reasoned answer consistent with 26 USC 152>",  
  "rejected": "<an incorrect or incomplete answer>"  
}
```

You must clearly document how you constructed the chosen and rejected responses (e.g., derived from the statute, from SMT outputs, or manually labeled). This documentation should appear in your report.

Part 2: Fine-Tune the Model

Base Model

Use **TinyLlama-1.1B** (or another comparably small open-source model approved for use on the HPC) as your base model. Specifying the base model ensures reproducibility across submissions and reduces memory and runtime variability.

Training Approach: LoRA or DPO (Your Choice)

You are required to apply **one** of the following post-training methods. You do not need to apply both.

Option	Description
Option A: LoRA	Fine-tune using Low-Rank Adaptation (LoRA) with 4-bit quantization (e.g., via <code>bitsandbytes</code>). Use your preference-formatted dataset in supervised fine-tuning mode.
Option B: DPO	Apply Direct Preference Optimization (e.g., using <code>DPOTrainer</code> from the <code>trl</code> library) using your preference-formatted dataset directly.

Note: These specific techniques were not covered in depth in lectures. This part of the project is intentionally open-ended to give you the opportunity to explore them independently and develop a practical understanding of post-training.

You must train **one model per dataset size** (small, medium, large) using your chosen method, for a total of **three fine-tuned models**.

Part 3: Evaluate on Fact Patterns

In legal terms, a **fact pattern** is a description of the important facts about a situation — who is involved, what happened, and any details that might matter legally.

Example: *"Emily is a 22-year-old full-time college student who lives with her parents. She does not earn enough money to support herself, and her parents provide more than half of her financial support during the year."*

Question: Is Emily a dependent under 26 U.S.C. § 152(a)-(d)?

Provided Fact Patterns

The six fact patterns below are provided for you. You should **not** include these in your training or DPO datasets.

1. Alex is 17 years old, lives with his aunt full-time, and his aunt pays all of his living expenses. His parents live in a different state and provide no financial support.
2. Maria is a 25-year-old graduate student who lives in university housing. Her parents pay her tuition but she covers all other living expenses through a graduate stipend.
3. James is 15 years old and lives with his parents during the school year, but spends summers with his grandparents. His parents claim him on their taxes and provide all of his financial support.
4. Linda is 45 years old, lives with her adult daughter, and earned \$5,000 this year from part-time work. Her daughter provides more than half of Linda's total support for the year.
5. Carlos is a 19-year-old who dropped out of college after one semester. He lives with his parents, works part-time earning \$8,000 per year, but his parents still provide the majority of his financial support.
6. Sophie is 16 years old, lives with a foster family, and has no contact with her biological parents. The foster family provides all of her financial support.

Evaluation Procedure

For **each** of the three models (small, medium, large), and for **each** of the six fact patterns:

1. Input the fact pattern into the fine-tuned model. Record the output.
 2. Classify the fact pattern using your SMT solver from Project 4 (or a revised version if needed). Record the SMT verdict.
 3. Compare the model output with the SMT verdict.
 4. Label each case as one of:
 - **Correct** — Model agrees with SMT
 - **Error** — Model output and SMT result do not match. Please check manually whether the issue lies in your SMT specification (and revise it if necessary) or in the model's output.
 - **Ambiguous** — The SMT result is indeterminate or multiple readings are possible.
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Deliverables

Please submit all of the following:

- Your dataset (all three sizes, in the required JSON format)
- Your training code or notebook
- Your SMT specification (from Project 4, revised if necessary)
- Six fact patterns with model outputs (from all three model sizes) and SMT verdicts
- A **summary table** for each model size, formatted as follows:

Summary Table (one per model size)

Fact Pattern	Model Output	SMT Verdict	Status
Case 1	Yes	Yes	Correct
Case 2	No	Yes	Error
Case 3	No	No	Correct
Case 4	Yes	No	Error
Case 5	Yes	Unknown	Ambiguous
Case 6	No	Unknown	Ambiguous

Report

Write a reflection of **no more than 500 words** addressing the following:

1. **Dataset ablation:** How did model performance change as dataset size increased from 10 to 20 to 50 examples? Did more data consistently help?
 2. **Method justification:** You chose either LoRA or DPO. Explain why you believe your chosen method is more appropriate for this task than the alternative. What are the tradeoffs?
 3. **Model behavior:** Describe your model's overall strengths and weaknesses in assessing dependency status. Were there systematic error patterns across the fact patterns?
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Submission Details

Please submit everything used to create your final deliverables and your report on Gradescope. We will need to execute your code on the HPC environment for grading and we want to be able to reproduce your results, so please let us know as early as possible if anything needs to be added. Please test your code on the HPC before making your final submission.

Please include a copy of your fact pattern summary tables (all three) in a separate PDF or text file.

If there are any special instructions needed to run your code — such as memory or hardware requirements — please include these at the very beginning of your report. If your code has dependencies not present on the HPC at the time of submission, we cannot guarantee they will be installed after you submit.

As with previous assignments, do not use libraries or AI models that require paid subscriptions.

Please only submit the files needed for your code and report. Do not include system files such as `.DS_Store` or `__MACOSX` files.